Research in Computing

Week – 1

Introduction & Module Overview

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Agenda

- Introduction
- Module Overview
- What is 'Research'?
- Research in Computing at NCI
- The Process of Research
- Resources

About me

About Me



- 2007 BSc Computer Science (GIFT University)
- 2009 MSc Ubiquitous Computing (University College Dublin)



- 2014 PhD in Computer Science (University College Dublin)
- 2015 Post-Doctoral Fellow in Computer Science
  The Insight SFI Research Centre for Data Analytics
  University of Galway, EU Funded Projects



- 2019 Senior Post-Doctoral Fellow in Computer Science, TCD
  Projects with Industry



- 2021 Research Fellow, Trinity College Dublin
- 2022 Adjunct Professor in Computer Science, Trinity College Dublin



- 2020 Associate Faculty in Computer Science, NCI

About the module

Module Descriptions

Learning Outcomes	
On successful completion of this module the learner will be able to:	
#	**Learning Outcome Description**
LO1	Understand what constitutes a good research question. Propose a research question and identify its implications with regard to the choice of subject;Critically assess and select methods for addressing the research question, including originality considerations;Propose research objectives and identify possible deliverables
LO2	Create a literature review which situates the work with regard to state of the art and seminal work. Develop a research and development design and methodology;
LO3	Demonstrate an ability to write a comprehensive research plan that explores research methods and deliverables for a specific subject in computing.Understand the need for reproducibility as a minimum standard for assessing the validity of the results of research
LO4	Understand the ethical issues that need to be addressed when conducting research;

http://courses.ncirl.ie/index.cfm/page/module/moduleId/67757

Module Content

Indicative Content
Research Questions and Literature Review Structure and purpose of a literature review. Search tools and sources. Selecting and coping with literature
Research Methodology, Research Questions and Literature Review Exploring different research methodologies and assessing the context for these research methodologies.. Formulating a research question.. Ethics in research.
Research in Computing The research community and their major platforms (journals, conferences, etc). . Making use of research articles to make informed choices in development
Research in Computing Planning software development and evaluation; User involvement; Descriptive, theory oriented and applied projects
Scientific Writing and Research Documentation Proposal structure. Selection and assessing the quality of literature.
Scientific Writing and Research Documentation Project structure. Citations and referencing.
Scientific Writing and Research Documentation Presenting qualitative data. Presenting quantitative data
Scientific Writing and Research Documentation The importance of ethics and reproducibility in research..
Scientific Writing and Research Documentation Scientific writing and style considerations.. Plagiarism and self-plagiarism
Technical Information Reading, understanding and summarising technical material, including source code, academic articles, patents, and documentation
Technical Information Writing effective technical documentation and materials.
Communication Dynamics of oral, written, and electronic team and group communication

Module Assessment

Assessments

Full Time

Coursework

Assessment Type:	Formative Assessment	**% of total:**	Non-Marked
Assessment Date:	n/a	**Outcome addressed:**	1,2,3,4
Non-Marked:	Yes		

Assessment Description:
n/a

Research Question
Due Week – 4

Assessment Type:	Continuous Assessment	**% of total:**	20
Assessment Date:	n/a	**Outcome addressed:**	1
Non-Marked:	No		

Assessment Description:
A written coursework assignment in which the student: • identifies an appropriate topic; • proposes a suitable research question; • list of objectives and identification of project beneficiaries; • Justifies the proposed research by citing 3-5 key sources in the domain; • explains the motivation for and expected contribution to knowledge of the proposed work; • addresses the feasibility and ethics of the proposed study;

Assessment Type:	Continuous Assessment	**% of total:**	80
Assessment Date:	n/a	**Outcome addressed:**	2,3,4
Non-Marked:	No		

Research Proposal
Due Week – 11

Assessment Description:
A written coursework assignment in consisting of: • an abstract – providing a summary of the proposal; • a literature review, comparing and contrasting prior work, leading to a clear research question and situating the proposed research question in context; • the proposed methodology and specification, including a project plan and all software deliverables; A 10 minute video presentation of the proposed research, giving an insight into the student's research interests.

Module Delivery

- Every Friday 09.00 – 11.00 (Lecture) S2.04 - Spencer Dock Level 2
- Every Friday 11.00 – 13.00 (Lab) S2.04 - Spencer Dock Level 2

Module Resources

Recommended Book Resources

Justin Zobel. (2015), Writing for Computer Science, Springer, p.284, [ISBN: 1447166388].

Christian W. Dawson. (2015), Projects in Computing and Information Systems, Prentice Hall, p.320, [ISBN: 1292073462].

John W. Creswell,J. David Creswell. Research Design, [ISBN: 1506386768].

Supplementary Book Resources

Gary Thomas. (2017), How to Do Your Research Project, Sage Publications Limited, p.360, [ISBN: 147394886X].

Justin Kitzes,Daniel Turek,Fatma Deniz. (2017), The Practice of Reproducible Research, Univ of California Press, p.368, [ISBN: 0520294750].

David Evans,Paul Gruba,Justin Zobel. (2014), How to Write a Better Thesis, Springer, p.167, [ISBN: 3319042858].

Diana Ridley. (2012), The Literature Review, SAGE Publications, p.214, [ISBN: 1446201430].

What is Research?

What is Research?

- **Merriam-Webster Dictionary**
  "A studious inquiry or examination; especially investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws.“
- **Oxford Concise Dictionary**
  - "The systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions.“
  - “A systematic effort to discover new facts or collate existing ones using scientific study or critical investigation.”
- **Philosophical View**
  "The art of asking questions that have useful answers that can be methodologically determined.“
- **Knowledge Creation Perspective**
  "Research is how we create knowledge."

Research is a **systematic process** of exploring and investigating questions to uncover facts, solve problems, and expand our understanding. It is also a **creative journey** to develop **new ideas** and apply them in meaningful ways.

Examples:

•**Cybersecurity**: Research on "What patterns in network traffic indicate a potential Distributed Denial of Service (DDoS) attack?"

•**Cloud Computing**: Investigating "What algorithms can predict cloud resource demand during peak usage?"

What is research?

- Research is what we do when we have a question or a problem we want to resolve
- We may already think we know the answer to our question already
- We may think the answer is obvious, common sense even
- But until we have subjected our problem to rigorous scientific scrutiny, our 'knowledge' remains little more than guesswork or at best, intuition.

Examples:
•**Cybersecurity**: Instead of assuming that an encryption method is secure, research might involve testing its resistance to brute-force attacks.
•**Cloud Computing**: Rather than guessing that a load-balancer is efficient, researchers analyze performance data to confirm its effectiveness.

What is research?

- First priority is to formulate your question
- Then figure out how you are going to answer it
  - How have others answered it?
  - How does your proposal fit in with what others have done?
  - How will you know when you have answered it?
- Then you can present your answer

Examples:
•**Cybersecurity**: Testing how AI models classify malware based on large datasets.
•**Cloud Computing**: Experimenting with real-time data to validate predictive scaling algorithms for virtual machines.

Scientific research

- It is the process of formulating specific questions and finding answers in order to understand a phenomenon better
- Systematic search for information
- What distinguish scientific research from other forms of research is the emphasis on using integrated **empirical** and **rational** processes
  - gaining information through sensory experiences and reasoning

- **Examples from Cybersecurity and Cloud Computing:**
- **Cybersecurity**:
  - *Question*: "What network traffic patterns indicate an ongoing DDoS attack?"
  - *Empirical*: Analyze traffic logs for anomalies (e.g., sudden spikes in packets from multiple IPs).
  - *Rational*: Use statistical models to predict whether the observed patterns are indicative of an attack.
- **Cloud Computing**:
  - *Question*: "How can predictive algorithms optimize resource allocation in cloud environments?"
  - *Empirical*: Collect usage data (e.g., CPU and memory loads).
  - *Rational*: Develop a mathematical model to predict peak usage and allocate resources dynamically.

Elements of research

- Facts
  - data or empirical observations
- Observation:
  - empirical process of recognising and recording facts
- Inference:
  - intellectual process in which conclusions are derived from facts or ideas
- Constructs
  - Non observable inferred events (gravity- electricity – intelligence, personality, user-experience)

- **Examples from Cybersecurity and Cloud Computing:**
- **Cybersecurity**:
  - *Facts*: Packet logs showing unusually high traffic from certain IPs.
  - *Observation*: Recording timestamps and frequency of incoming packets.
  - *Inference*: Concluding that it could be a DDoS attack based on traffic behavior.
  - *Constructs*: The inferred "severity" of the attack based on the duration and intensity of the traffic.
- **Cloud Computing**:
  - *Facts*: Historical CPU and memory usage data.
  - *Observation*: Identifying peak times when usage spikes.
  - *Inference*: Predicting future demand and scheduling resources accordingly.
  - *Constructs*: User satisfaction inferred from reduced latency during peak hours.

How Research in Cybersecurity looks like?

Key Focus Areas:

Threat Detection: Researching ways to identify threats like malware, phishing, or ransomware.

Data Security in the Cloud: Exploring encryption methods or access control mechanisms to secure cloud data.

System Vulnerability Analysis: Investigating and fixing vulnerabilities in systems, networks, or cloud architectures.



Example:

- *Problem*: "How to detect ransomware spreading across cloud-based collaborative platforms?"
- *Solution*: Develop an anomaly detection algorithm that analyzes user behavior patterns in shared cloud environments.

Outcome Of Research

- New knowledge . . .
  - Student projects are research training, but most should ideally lead to publishable research outcomes, e.g., refereed journal or conference papers.
- New skills: communication, critical thinking, independence.

- **Examples**:
- **Cybersecurity**: Research leading to a lightweight intrusion detection system that uses fewer computational resources.
- **Cloud Computing**: Development of an algorithm that dynamically optimizes server workloads.

Purpose of Research

- Prepare you to produce a strong research project and report.
- Specifically,
  - How to read and understand research papers; and what to say about them
  - How to write about your own research, and how to present it to others
  - Library skills for researchers
  - Time planning and scheduling
  - How to design experiments, human studies, data, and study design
  - Ethics and research conduct

- **Examples**:
- **Cybersecurity**: Learning to read and analyze academic papers about AI-based threat detection.
- **Cloud Computing**: Designing experiments to test the scalability of cloud services under heavy loads.

Process of Doing Research

- Classical scientific method:
- Observation of some phenomenon
  - Begin by describing how research starts with observing a problem or situation.
  - Maybe systematic, occasional or accidental
- Some idea of an explanation (forming hypothesis)
  - Explain how researchers create possible explanations or predictions.
  - Usually informed by related work
- Testing of the hypothesis
  - Test and revision cycle

- **Examples**:
- **Cybersecurity**:
  - Observation: "Phishing emails are increasing in frequency."
  - Hypothesis: "AI-based email filters can reduce phishing by 90%."
  - Testing: Use datasets of emails to train and test an AI model for filtering.
- **Cloud Computing**:
  - Observation: "Server response times increase during high traffic."
  - Hypothesis: "Load balancing reduces response time by 30%."
  - Testing: Simulate traffic to measure the impact of load-balancing algorithms.

Hypothesis

- Probability of research
  - Nothing is certain
  - *The exception that “proves” the rule*
  - Scientific “truth” is usually a statement of what is most probable given the currently known data ...
  - ... within the given framework
- Statistical techniques try to help us show extent to which our results really do support the hypothesis
- A hypothesis makes a prediction of the expected outcome in a given situation
- The hypothesis is tested in an **experiment**

- **Examples**:
- **Cybersecurity**:
  - Hypothesis: "Introducing real-time anomaly detection will reduce network breaches."
  - Test: Deploy anomaly detection systems in a test network and analyze the results.
- **Cloud Computing**:
  - Hypothesis: "Predictive scaling will save 20% of cloud resource costs during peak hours."
  - Test: Implement predictive scaling in a cloud environment and compare cost savings.

Process of Doing Research

1. Identification of research problem (questions & hypotheses)
2. Conduct a literature review to see what has been done before.
3. Develop methods for addressing the problem.
4. Data collection, experimentation, mathematical proof
5. Analysis, interpretation, and critical evaluation
6. Write up & presentation

- **Examples**:
- **Cybersecurity**:
  - Research Problem: "How can IoT devices be secured from malware?"
  - Literature Review: Identify existing IoT security techniques.
  - Method: Test lightweight encryption algorithms.
  - Data Collection: Use IoT traffic data to evaluate encryption performance.
- **Cloud Computing**:
  - Research Problem: "How can we reduce energy consumption in data centers?"
  - Literature Review: Analyze existing energy optimization methods.
  - Method: Implement and test an energy-efficient algorithm.
  - Data Collection: Measure energy usage before and after optimization.

The Process Of Research

- **Speculate** Develop a concept, follow a train of thought, have a hunch that an idea is worthwhile.
- **Encapsulate** Form a specific research question.
- **Observe** Gather preliminary data, make an initial implementation, sketch out some proofs.
- **Hypothesis** Make testable predictions based on a strong understanding of the research question.
- **Test** Design instruments, undertake mathematical analysis, systematically gather data, perform robust experiments.
- **Analyse** Use critical argument to link the results to the hypothesis, thus demonstrating whether it is supported or contradicted.
- **Write-up** Use the results and analysis to convince skeptical examiners that your conclusions are interesting and correct.

- **Examples**:
- **Cybersecurity**:
  - Speculate: What causes frequent ransomware attacks?
  - Encapsulate: Formulate a research question like, "How can AI detect ransomware early?"
  - Test: Experiment with machine learning algorithms on ransomware datasets.
  - Analyze: Compare algorithm accuracy and runtime.
- **Cloud Computing**:
  - Speculate: Why do some servers overuse energy?
  - Encapsulate: Research question: "What load-balancing methods improve energy efficiency?"
  - Test: Simulate different methods under peak load conditions.

Aims Of Research

- To have recognition
- To have impact
- To create knowledge
- To create wealth
- To be part of something important
- To have fun
- To learn
- To propose
- To develop

- **Examples**:
- **Cybersecurity**: Research on lightweight cryptographic algorithms to secure IoT devices can create industry-wide impact.
- **Cloud Computing**: Developing innovative resource allocation methods can improve efficiency and reduce costs globally.

Acquiring knowledge

- Tenacity
  - Accept ideas as valid because they have been accepted for so long
- Intuition
  - Accepting ideas as valid because they ‘feel’ intuitively true
- Authority
  - Accepting ideas as valid because some respected authority asserts that the ideas are true
- Rationalism
  - Developing valid ideas using existing ideas and principles of logic
- Empiricism
  - Gaining knowledge through observations
- Science
  - A process which combines the principles of **rationalism** with the process of **empiricism**, using **rationalism** to **develop theories** and **empiricism to test these theories**

- **Examples**:
- **Cybersecurity**:
  - Empiricism: Collecting real-world attack data for analysis.
  - Rationalism: Designing an algorithm based on logical principles.
- **Cloud Computing**:
  - Empiricism: Observing resource utilization patterns.
  - Rationalism: Developing optimization models based on observed patterns.

Perspectives on research

- The traditional sciences were, **historically, *observational*** and concerned with quantifying and explaining the properties of the natural universe.
- **Much technical research today is applied** and concerns utility, that is, solution of problems in a practical context.
- But it **is not sufficient to just build** or define something that seems to solve a problem –research typically **involves understanding the properties of the solution**:
  - How far short of the ideal?
  - Scope, limitations?
  - Measurable properties?
  - Lessons for other problems?

- **Examples**:
- **Cybersecurity**:

Observational research might involve studying attack patterns, while applied research develops detection systems.

- **Cloud Computing**:

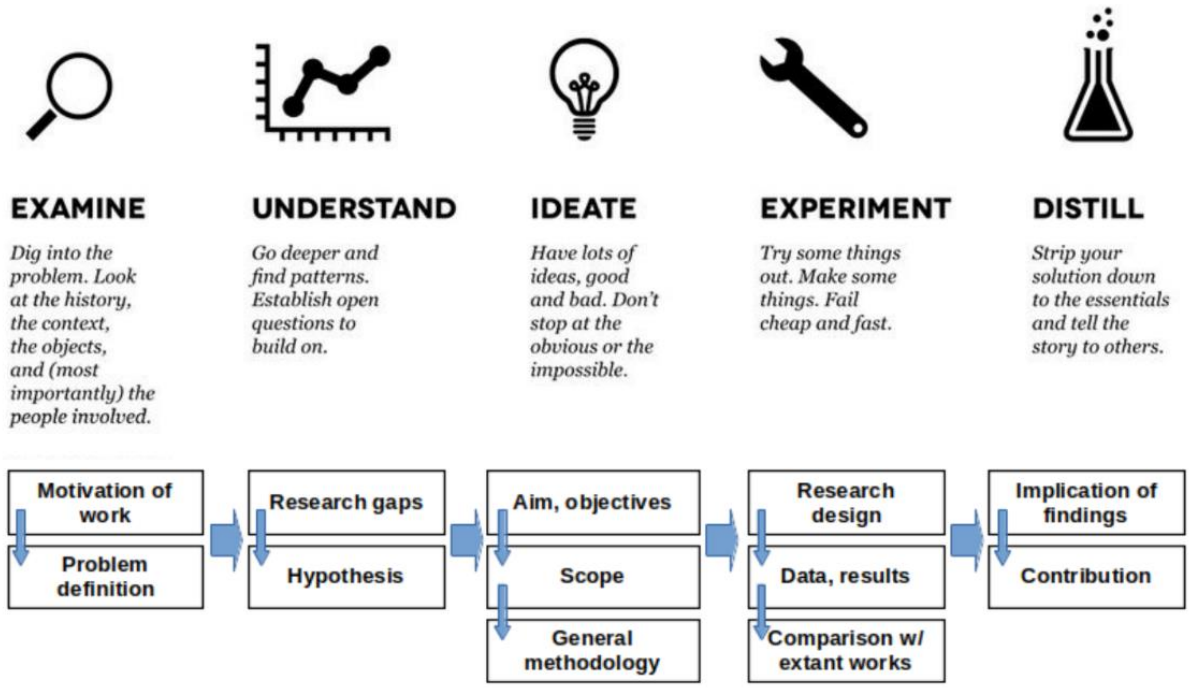
Observational research could measure energy use in servers, while applied research creates energy-efficient systems.

The major steps

- Find a research problem
- Justify why the problem is important
- Present the previous solutions to that problem – state of the art
- Discuss their advantages and disadvantages – literature review
- Motivate why a new solution is needed
- Describe your new solution
- Show that your solution has some advantage (previously unseen results)
- Discuss conclusions and future work

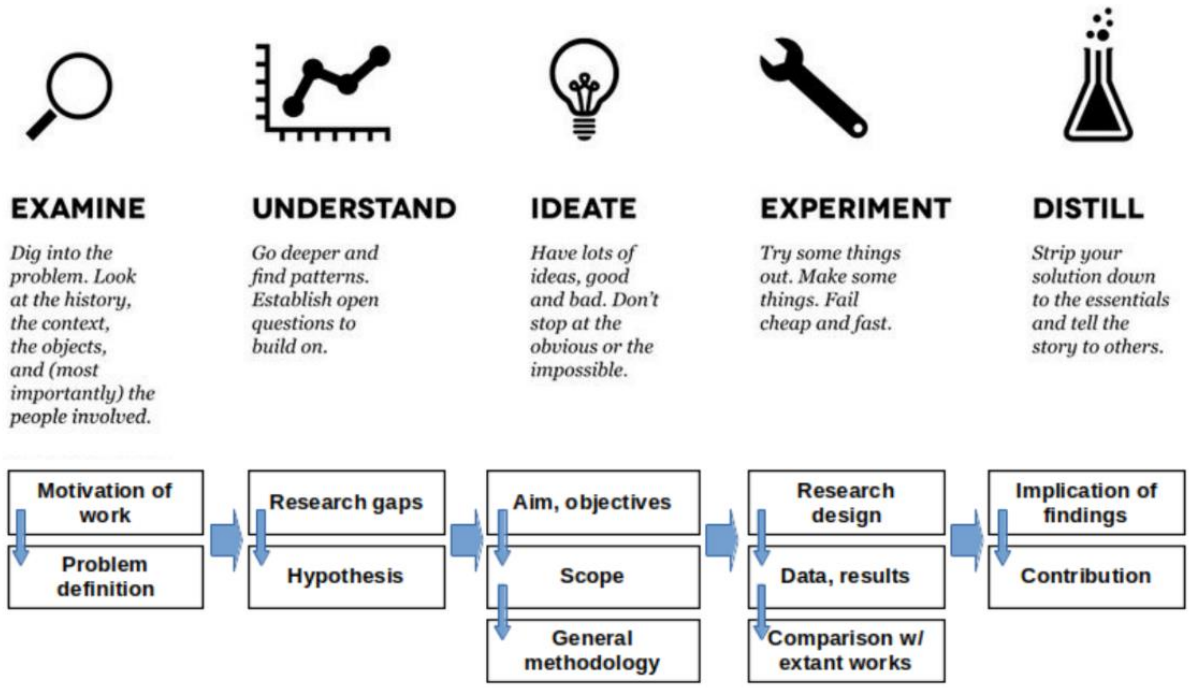
- **Examples**:
- **Cybersecurity**:
  - Problem: Frequent DDoS attacks.
  - Solution: Develop a machine learning model to detect attacks in real-time.
  - Test: Deploy the model in a live network and evaluate.
- **Cloud Computing**:
  - Problem: High energy consumption during server peaks.
  - Solution: Create a load-balancing algorithm.
  - Test: Compare energy usage before and after implementation.

Linking research phases to your MSC project content



- **Example from Cybersecurity:**
- **Examine**:
  - *Problem*: Increasing phishing attacks on enterprise email systems.
  - *Motivation*: Significant financial losses due to successful phishing.
- **Understand**:
  - Identify gaps: Lack of adaptive AI-based phishing detection systems.
  - Form hypothesis: "Using contextual data improves phishing detection rates."
- **Ideate**:
  - Objectives: Develop and test an AI-powered phishing email classifier.
  - Methodology: Use supervised learning with labeled email datasets.
- **Experiment**:
  - Design experiments: Test the model on datasets and compare results with existing methods.
- **Distill**:
  - Findings: AI classifier outperforms traditional rule-based filters.
  - Contribution: A new, adaptive phishing detection model.

Linking research phases to your MSC project content (Cont'd)



- **Example from Cloud Computing:**
- **Examine**:
  - *Problem*: Inefficient resource allocation during peak traffic times.
  - *Motivation*: Optimize costs and improve system performance.
- **Understand**:
  - Identify gaps: Existing load balancers lack predictive capabilities.
  - Form hypothesis: "Predictive analytics reduces cloud server downtime by 20%."
- **Ideate**:
  - Objectives: Develop a predictive load-balancing algorithm.
  - Scope: Focus on high-traffic e-commerce servers.
- **Experiment**:
  - Collect real-world traffic data, implement the algorithm, and measure results.
- **Distill**:
  - Findings: Predictive load balancing improved performance metrics.
  - Contribution: A scalable and efficient resource allocation method.